

Probability 1 – Exercises

Tutorial no. 11

27th Nov 2025

- 11.1** The random variables X and Y have a jointly absolutely continuous distribution, the marginal density function of X is $f_X(x) = \lambda^2 x e^{-\lambda x}$ if $x > 0$ and 0 if $x < 0$. The conditional distribution of Y given the value of X is uniform on the interval $[0, X]$. Calculate
- (a) the joint probability density function of X and Y ,
 - (b) the marginal density function of Y and the expectation $\mathbb{E}(Y)$,
 - (c) $\mathbb{E}(Y | X = x)$ and $\mathbb{E}(X | Y = y)$,
 - (d) $\mathbb{E}(X)$ (at this point it is simpler if we use the previous sub-exercise), and
 - (e) $\text{Cov}(X, Y)$.
- 11.2** We throw a coin 10 times, with a probability p of heads and probability $(1 - p)$ of tails at each throw. Let X be the number of *pure runs* (e.g. in TTHTT, $X = 3$). Calculate $\mathbb{E}(X)$ and $\text{Var}(X)$. (Hint: write X as a sum of some indicator variables of some pretty simple events.)
- 11.3** We throw with a die n times. Let X be the number of times we throw 1, and Y be the number of times we throw 2. What is the correlation of the two random variables?
- HW 11.4** (3 points) There are 37 pockets on the roulette wheel, from 0 to 36. Xavier always bets that the result is at least 19. Yvette always bets that the result is $1 \bmod 3$ (so, $1, 4, 7, \dots, 34$). Let's spin the wheel 20 times, independently. Let X be the number of times Xavier wins, and Y be the number of times Yvette wins. (It's possible that they both win a round, or that neither wins a round). What is the correlation of X and Y ?
- HW 11.5** (3 points) In an urn, there are 9 pink and 13 orange balls. We take out all balls without replacement. Let X be the number of *pure runs*. (Similarly to question 11.2, but with pink and orange instead of heads and tails). Calculate $\mathbb{E}(X)$.
- 11.6** Twelve people get on an elevator. They all choose a destination independently out of the 10 possible floors. Determine the expected value and the variance of the number of floors where the elevator stops.
- 11.7** Let (X, Y) has jointly uniform distribution on the triangle determined by the points $(-1, 0)$, $(0, 0)$, $(0, 2)$. (a) What is the two dimensional covariance matrix of (X, Y) ?
(b) Let $Z = 7X + 2Y$. What is the two dimensional covariance matrix of (X, Z) ?
- HW 11.8** (4 points) Let X and Y be random variables that can only take two values: $X \in \{x_1, x_2\}$ and $Y \in \{y_1, y_2\}$. Prove that if $\text{Cov}(X, Y) = 0$, then they are independent. (Not true in general!)
- 11.9** Let (X, Y) be the coordinates of a uniformly chosen point on the circumference of the circle centered at $(1, 1)$ with radius 1. $\text{Cov}(X, Y) = ?$